

Global CO₂ and Climate Change

**A deeper look at the carbon dioxide cycle, greenhouse gases,
and oceanic processes over the last 200 years**

OCN 623 – Chemical Oceanography

Reading: Emerson & Hedges Ch. 11

Student Learning Outcomes (SLOs)

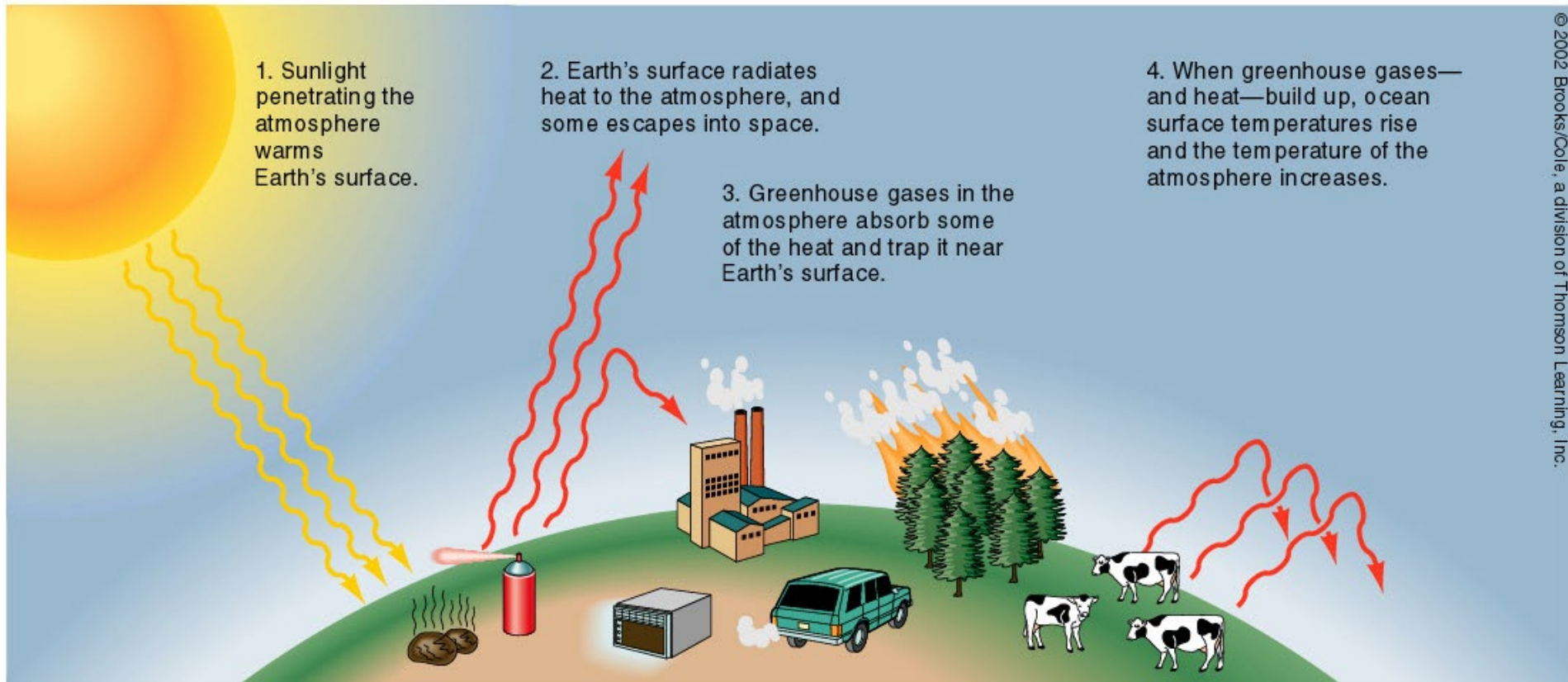
At the completion of today's section, students should be able to:

1. Know what affects CO₂ variability in the atmosphere
2. Describe how has global carbon cycle changed over the past two centuries
3. Describe the long-term fate of anthropogenic carbon
4. Identify the role of the ocean in climate change
5. implications of climate change on society

How does CO₂ affect climate?

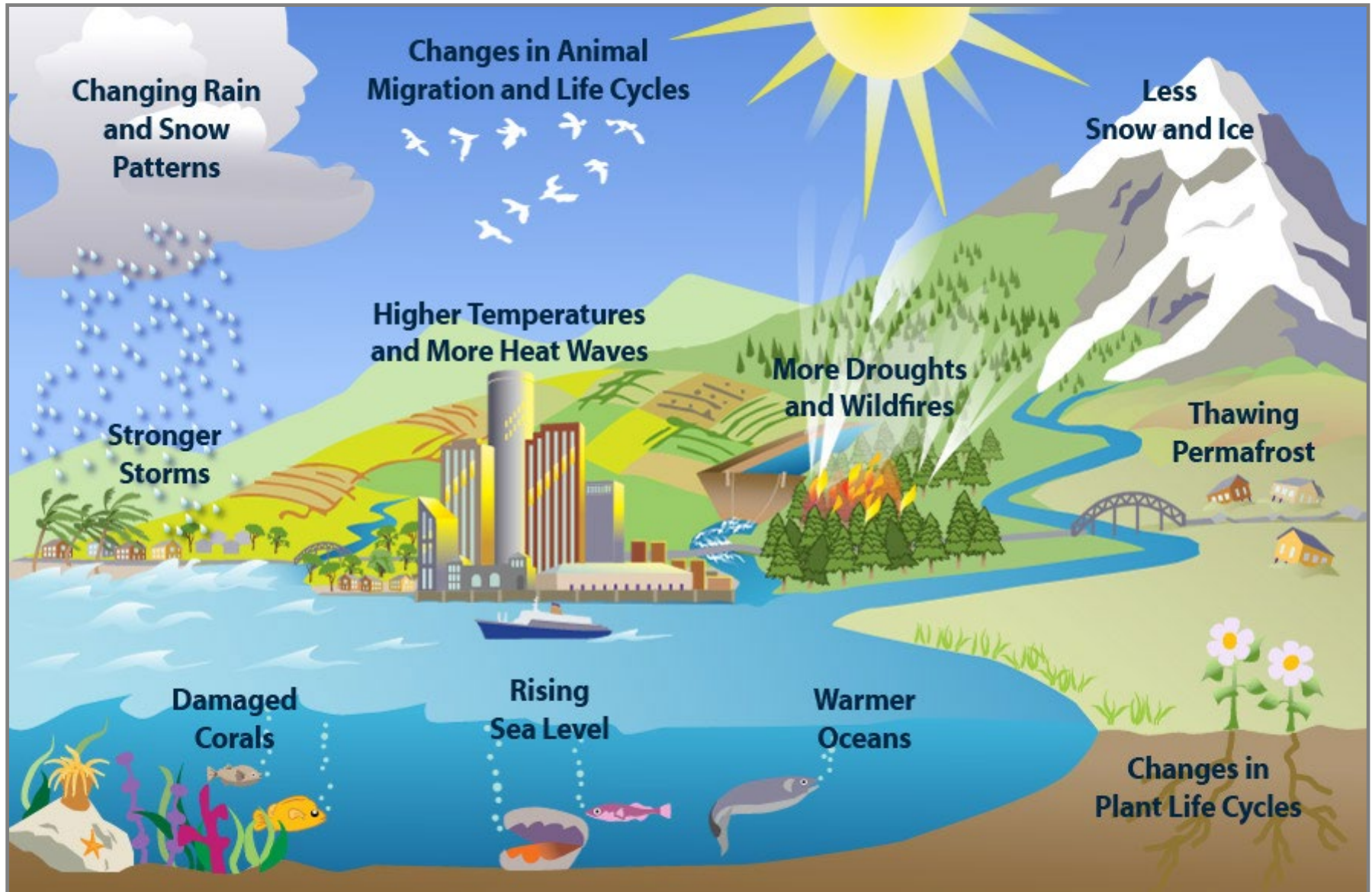
The Earth's temperature is set by a *Radiation Balance*:

If more **heat arrives from the sun** than can **escape as infrared** (IR) rays, the Earth gets warmer.



CO₂ and other greenhouse gases absorb IR, so an increase in CO₂ causes an increase in temperature.

Climate Change versus Global Warming

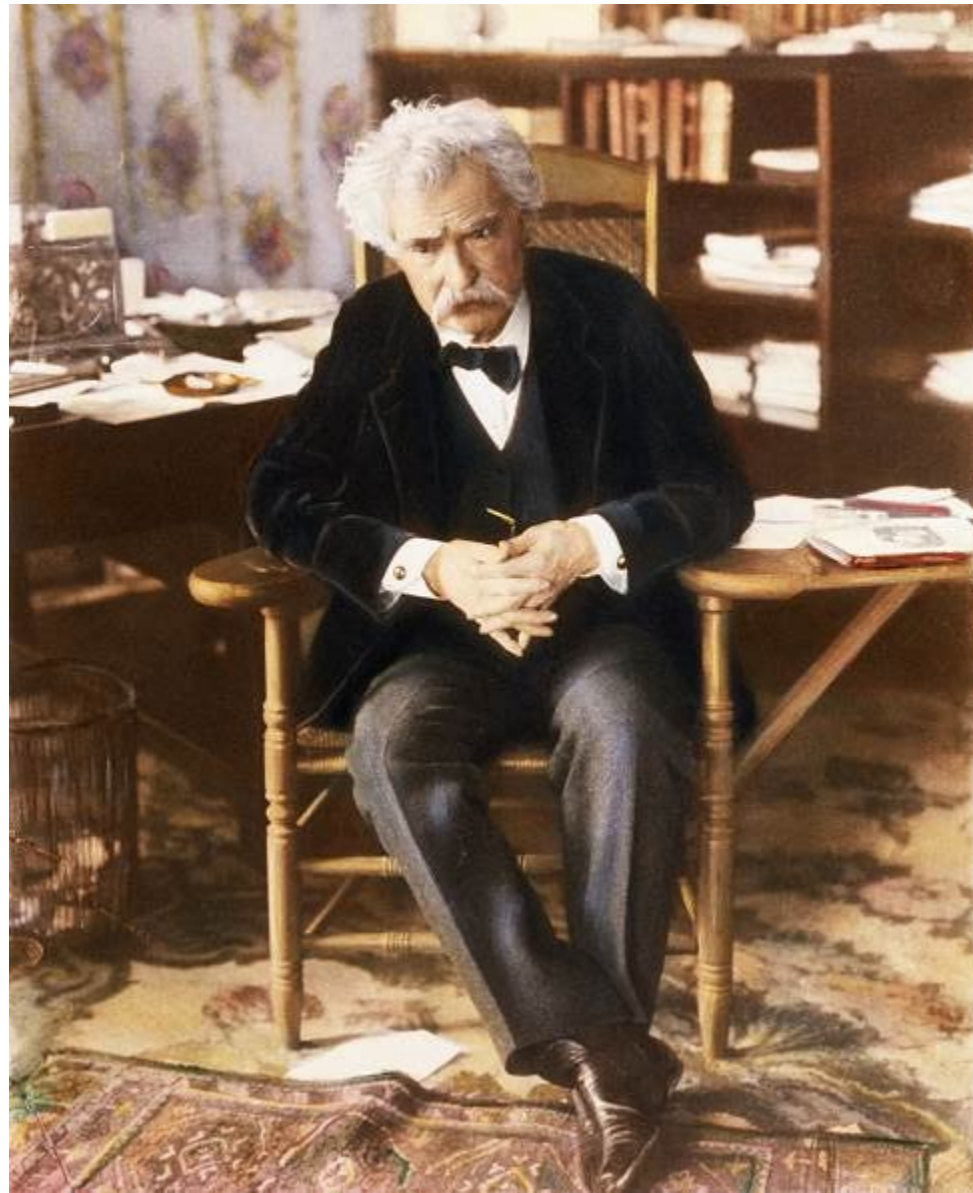


Webster:

“Climate: the average condition of the weather at a place over a period of years as exhibited by temperature, wind velocity, and precipitation”

Mark Twain:

Climate is what you expect,
weather is what you get.

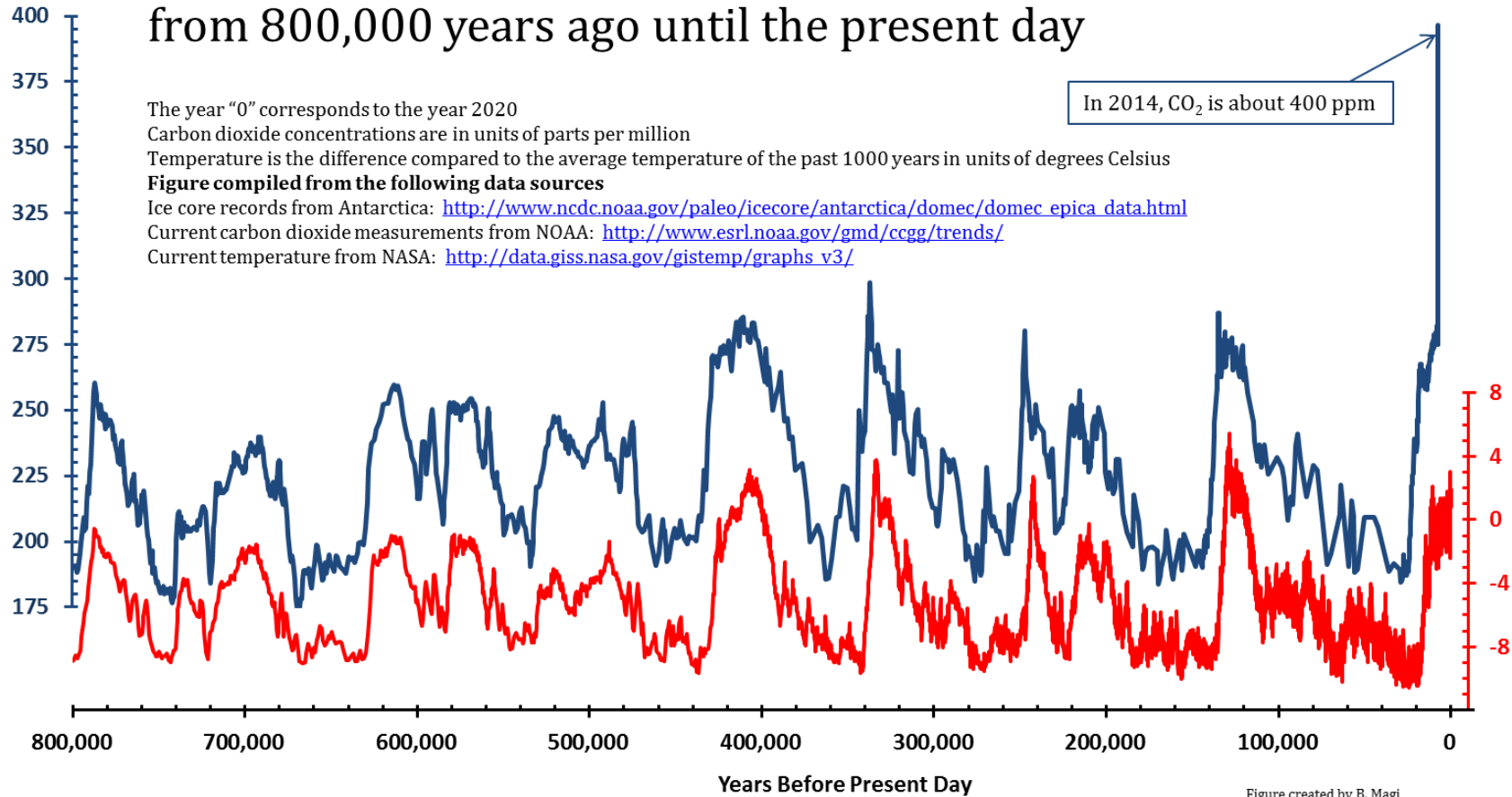


- Climate change is a natural process that has happened for billions of years
- Human activities are changing the rate of climate change
- Climate change includes many more effects than warming

Are global temperatures linked to atmospheric CO₂ ?

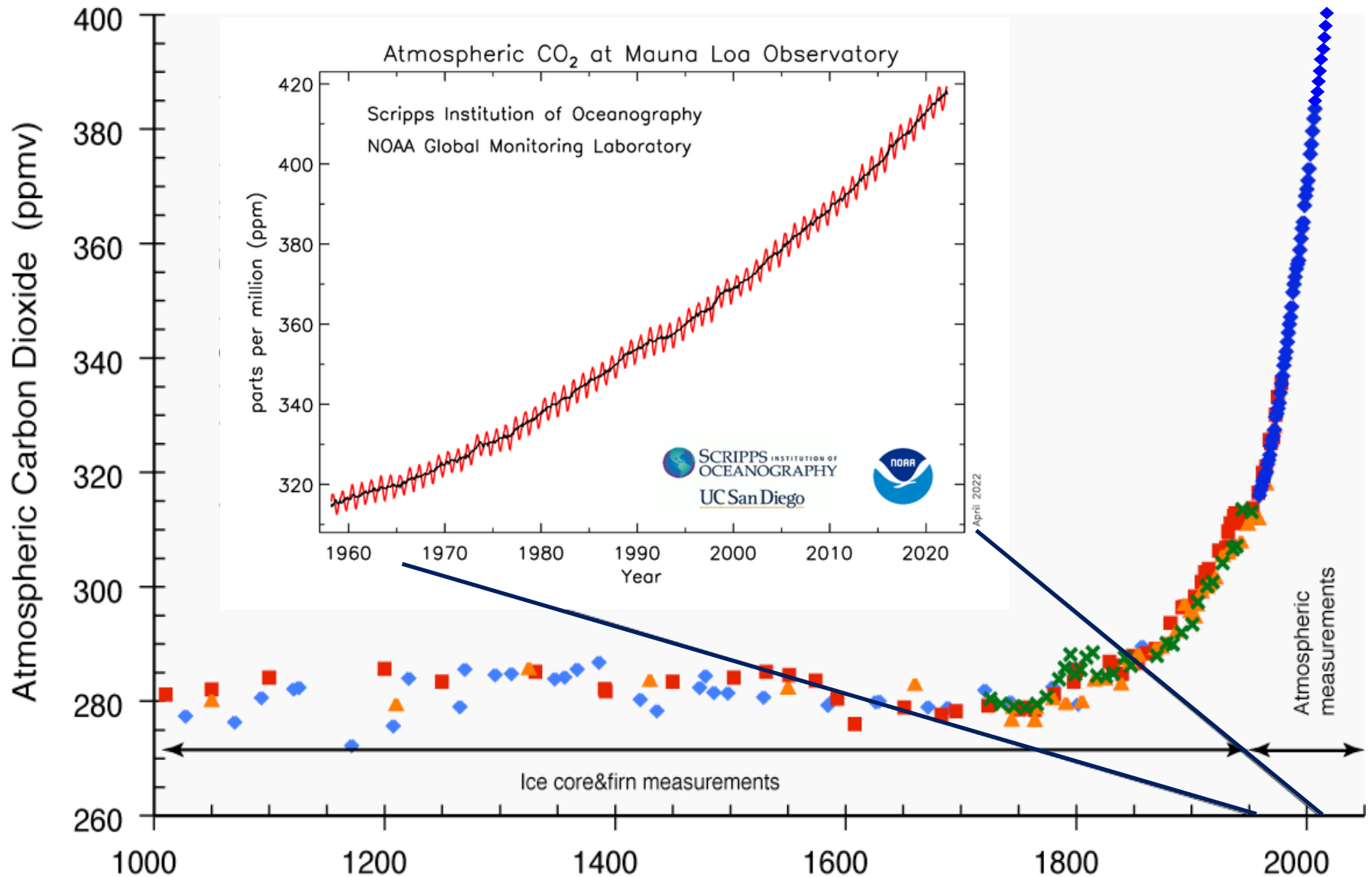
Vostok Ice Core CO₂ Concentration and Temperature Variation Record

Carbon dioxide and the **temperature of our planet** from 800,000 years ago until the present day



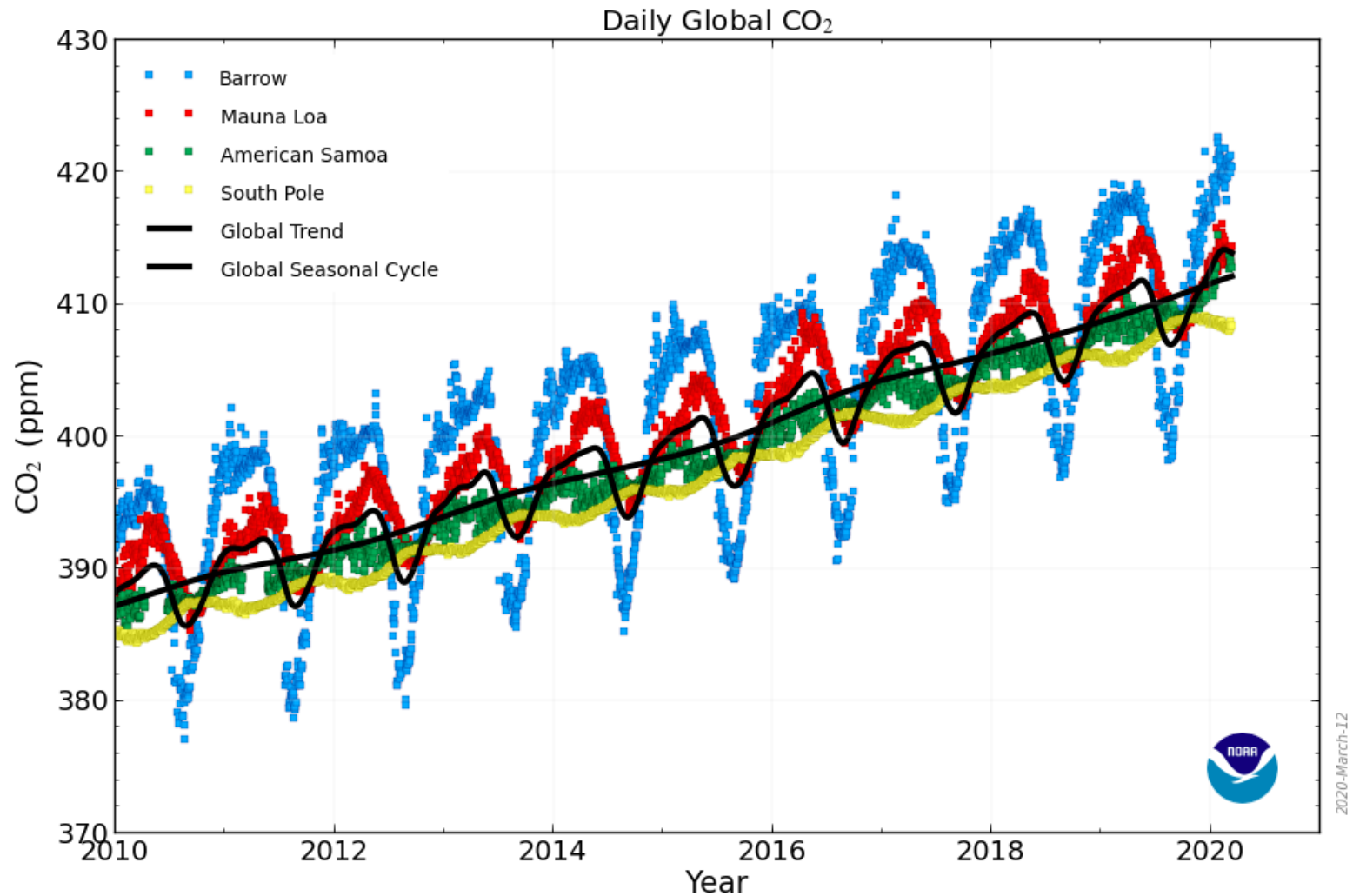
IPCC: "Global mean surface temperature has increased more than 0.5°C since the beginning of the 20th century, with this warming likely being the largest during any century over the past 1,000 years for the Northern hemisphere."

Carbon Dioxide Levels Continue to Rise in the Atmosphere



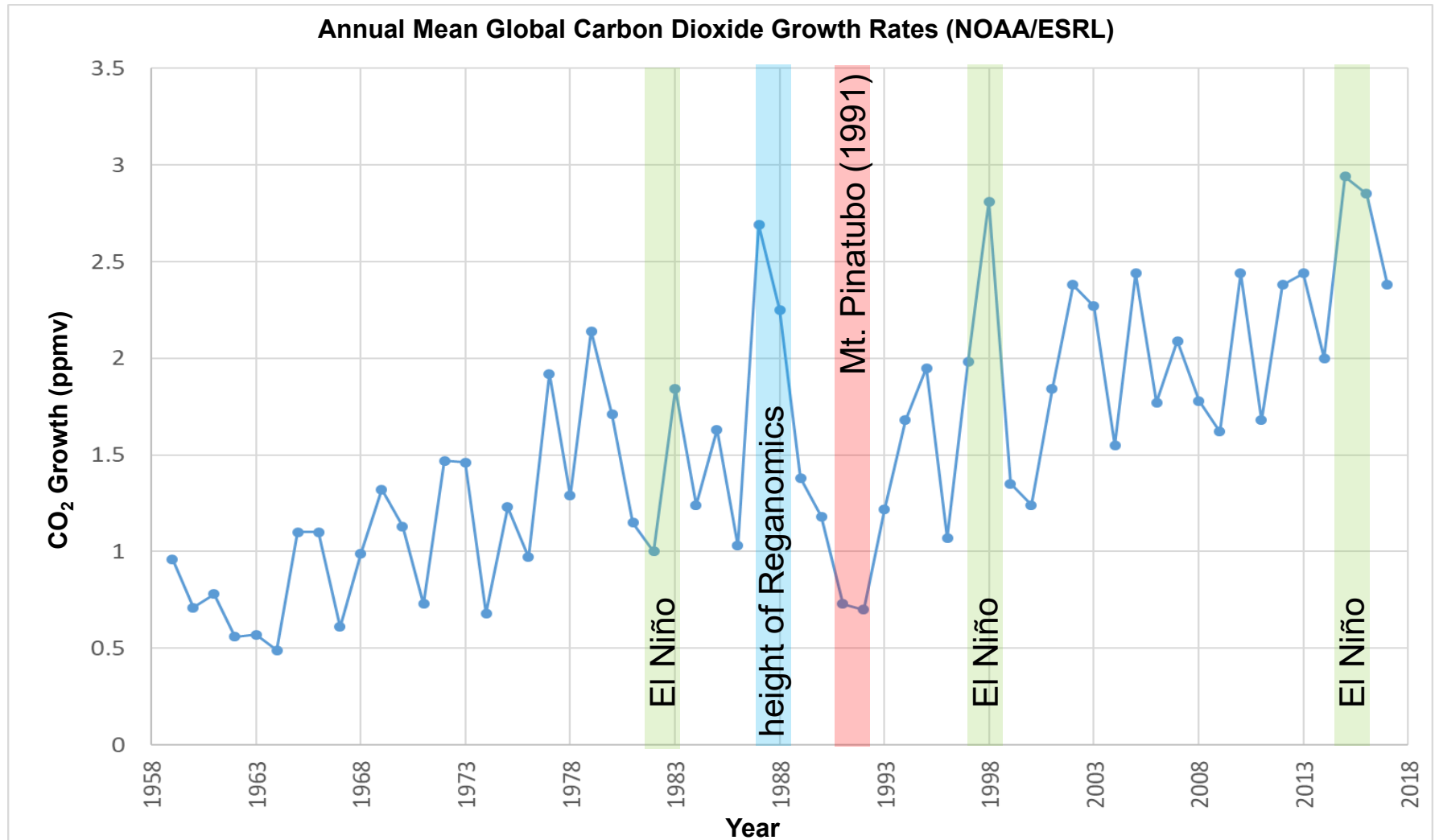
Adapted from Sarmiento and Gruber 2002 using Trends online data

Recent atmospheric carbon dioxide levels



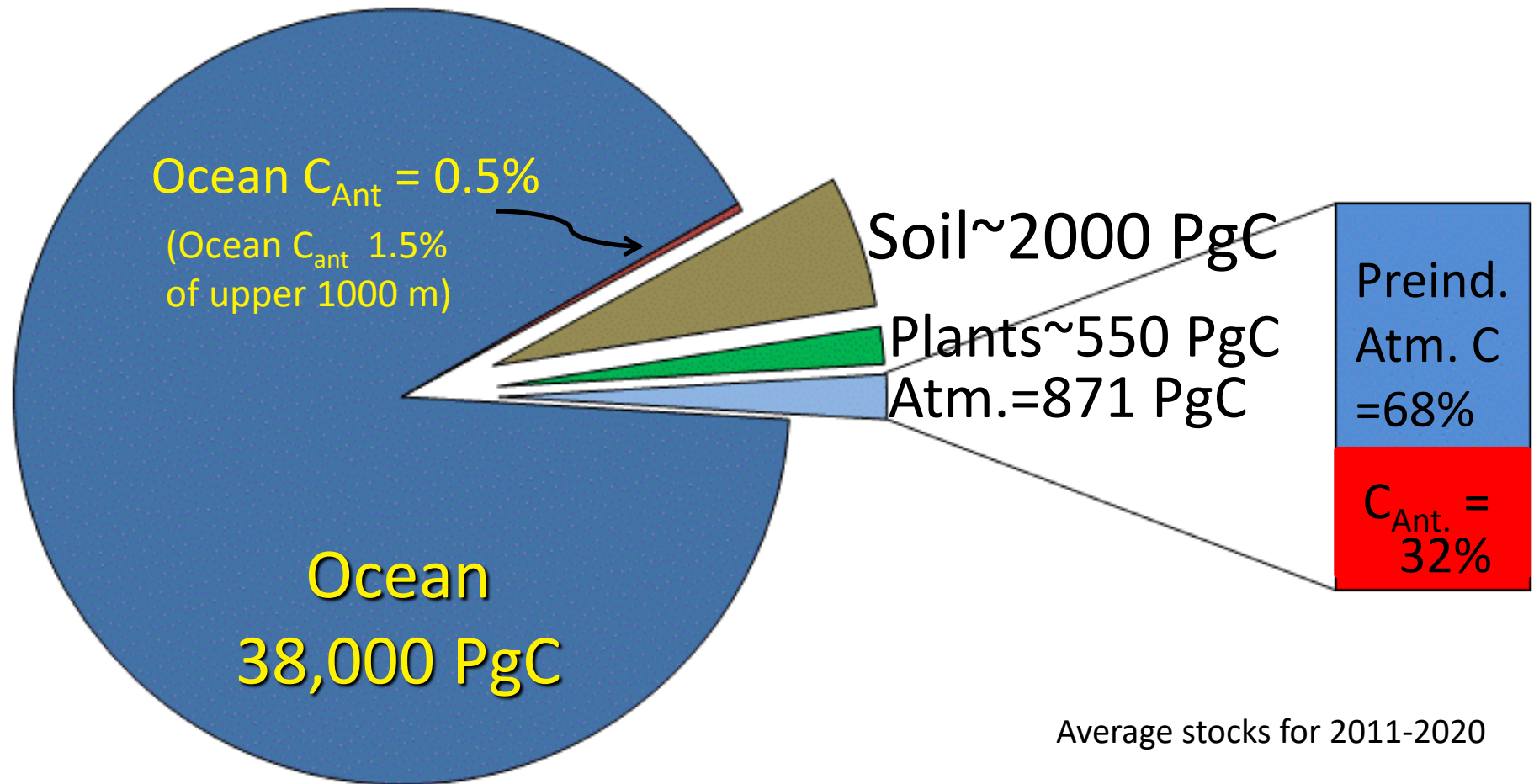
Atmospheric CO₂ levels have risen from ~315 ppmv in 1958 to 416 ppmv in 2021 (~24%)

Rate of increase of atmospheric CO₂ is not constant



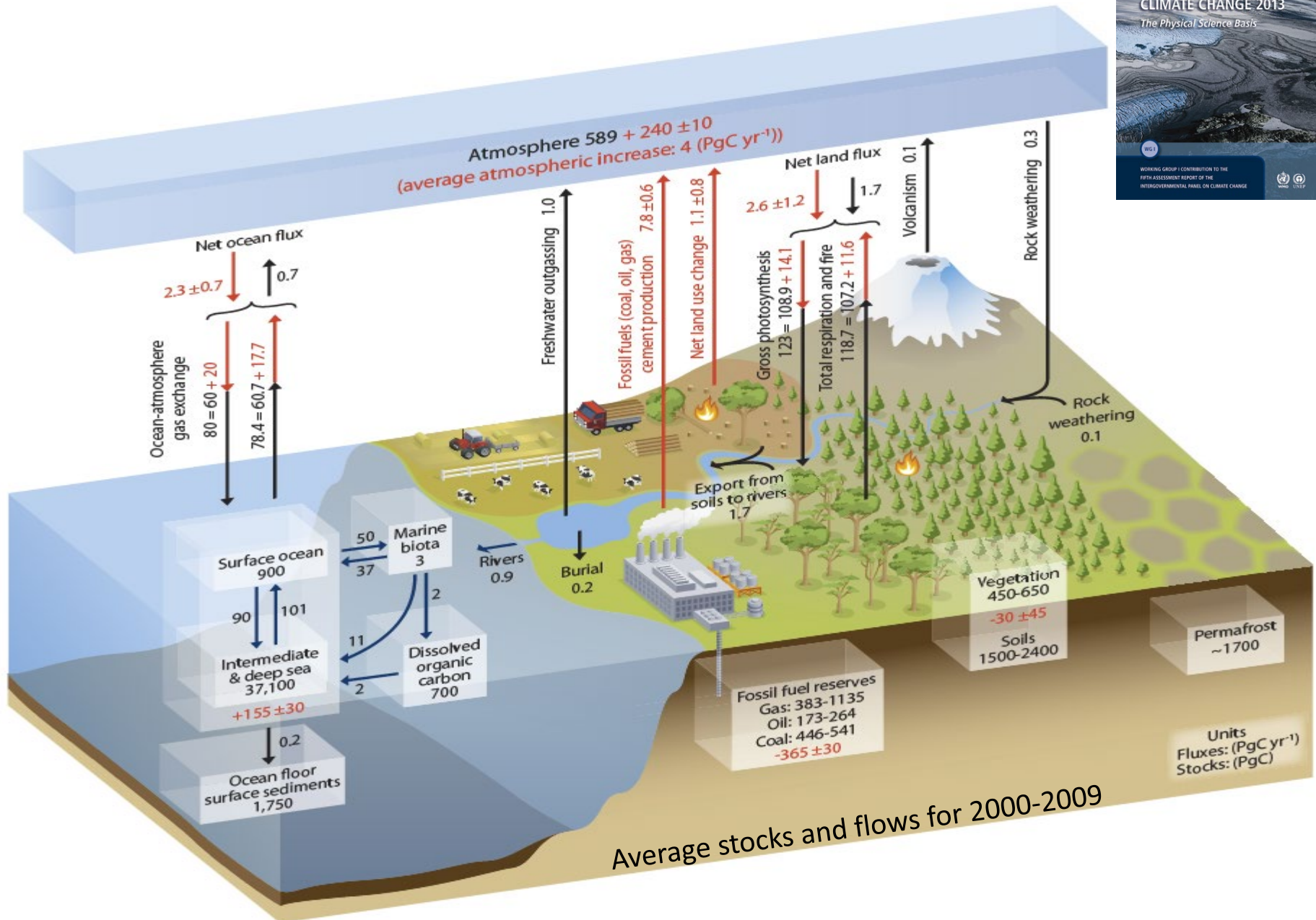
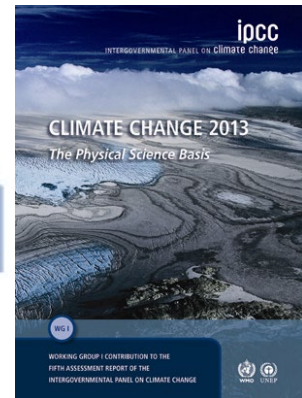
Varies with: Economic activity and natural sinks/sources
(El Nino, droughts, fires, Volcanic activity)

Carbon Inventories of Reservoirs that Naturally Exchange Carbon on Time Scales of Decades to Centuries

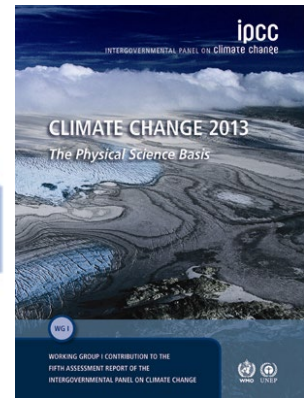
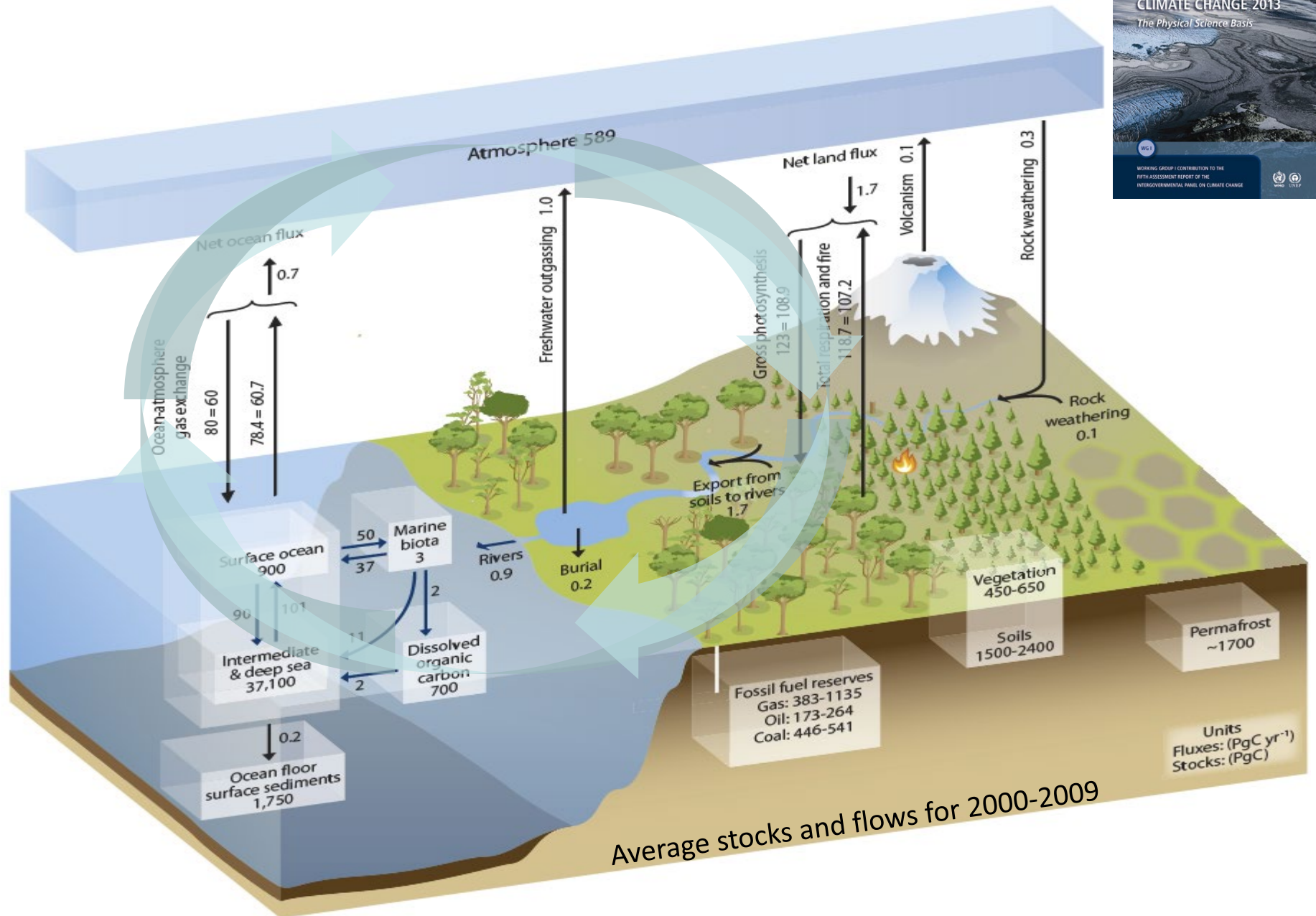


- Oceans contain ~90% of carbon in this 4 component system
- anthropogenic component is difficult to detect in the ocean

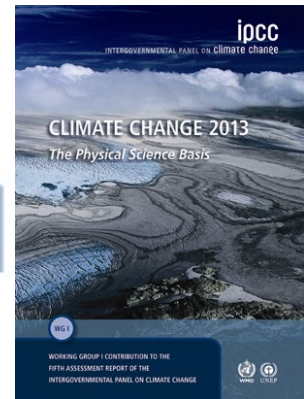
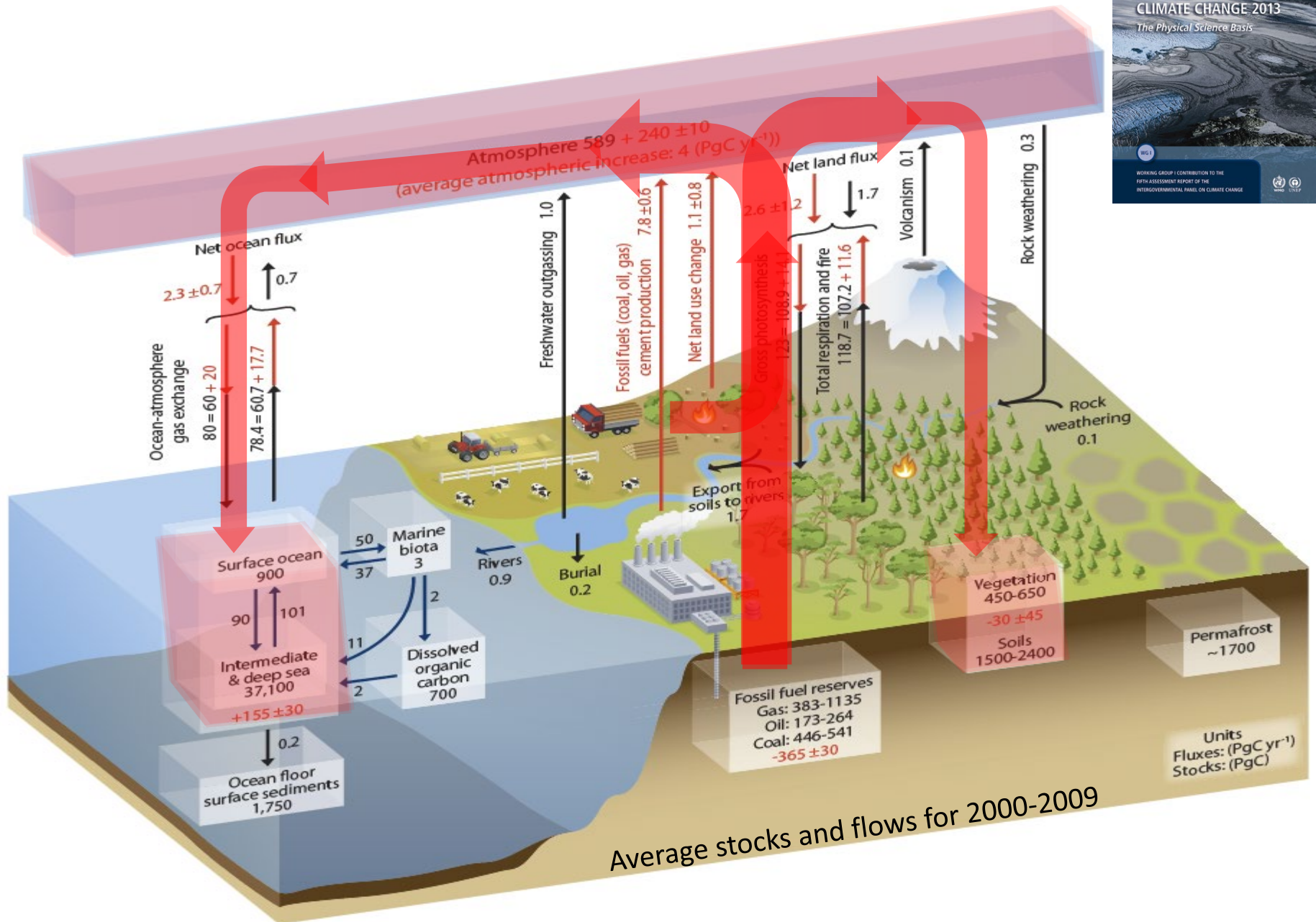
IPCC global carbon cycle and the flows of carbon



Preindustrial C cycle was from the Ocean to the Land



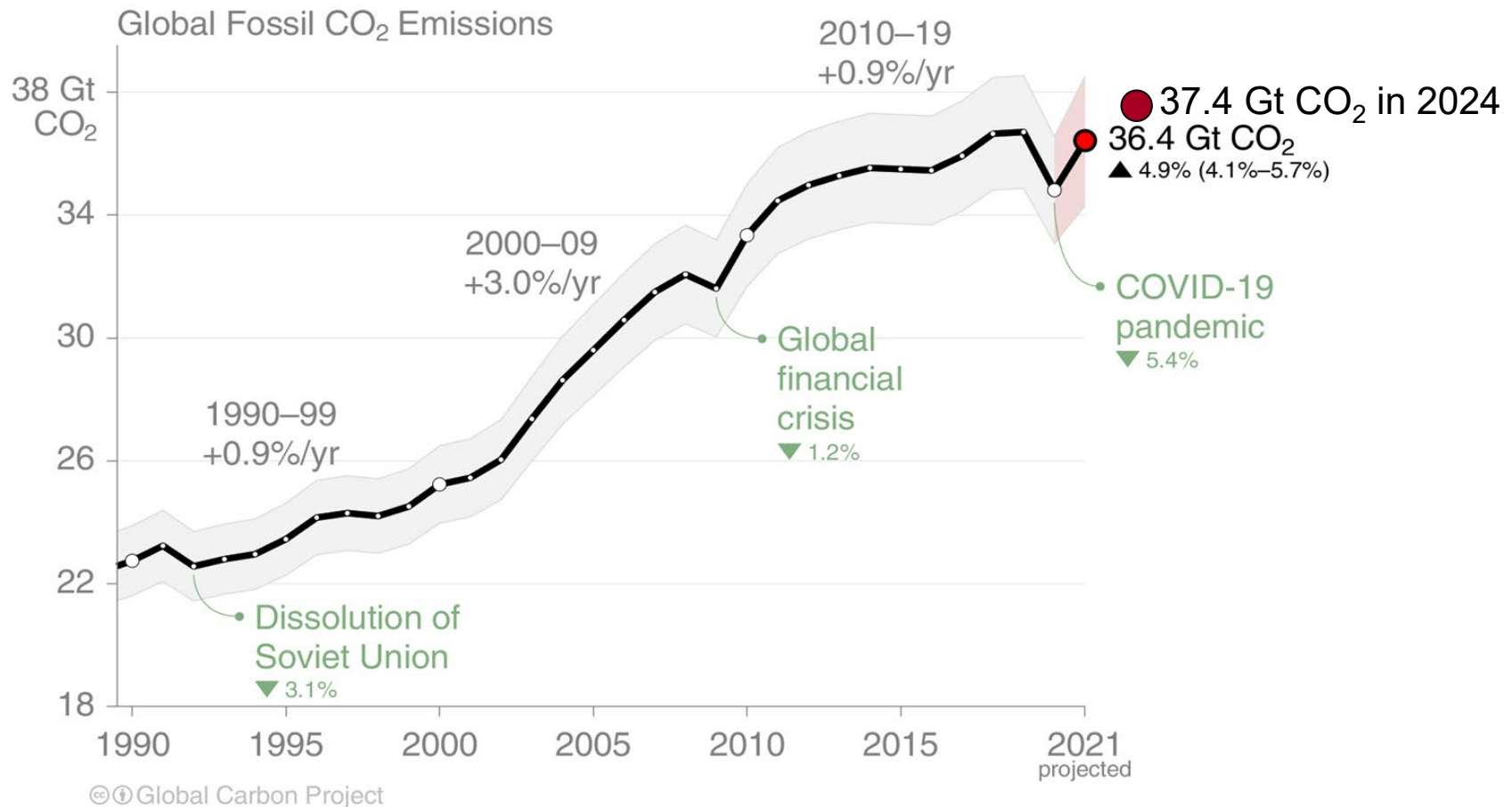
Today we have reversed that flow



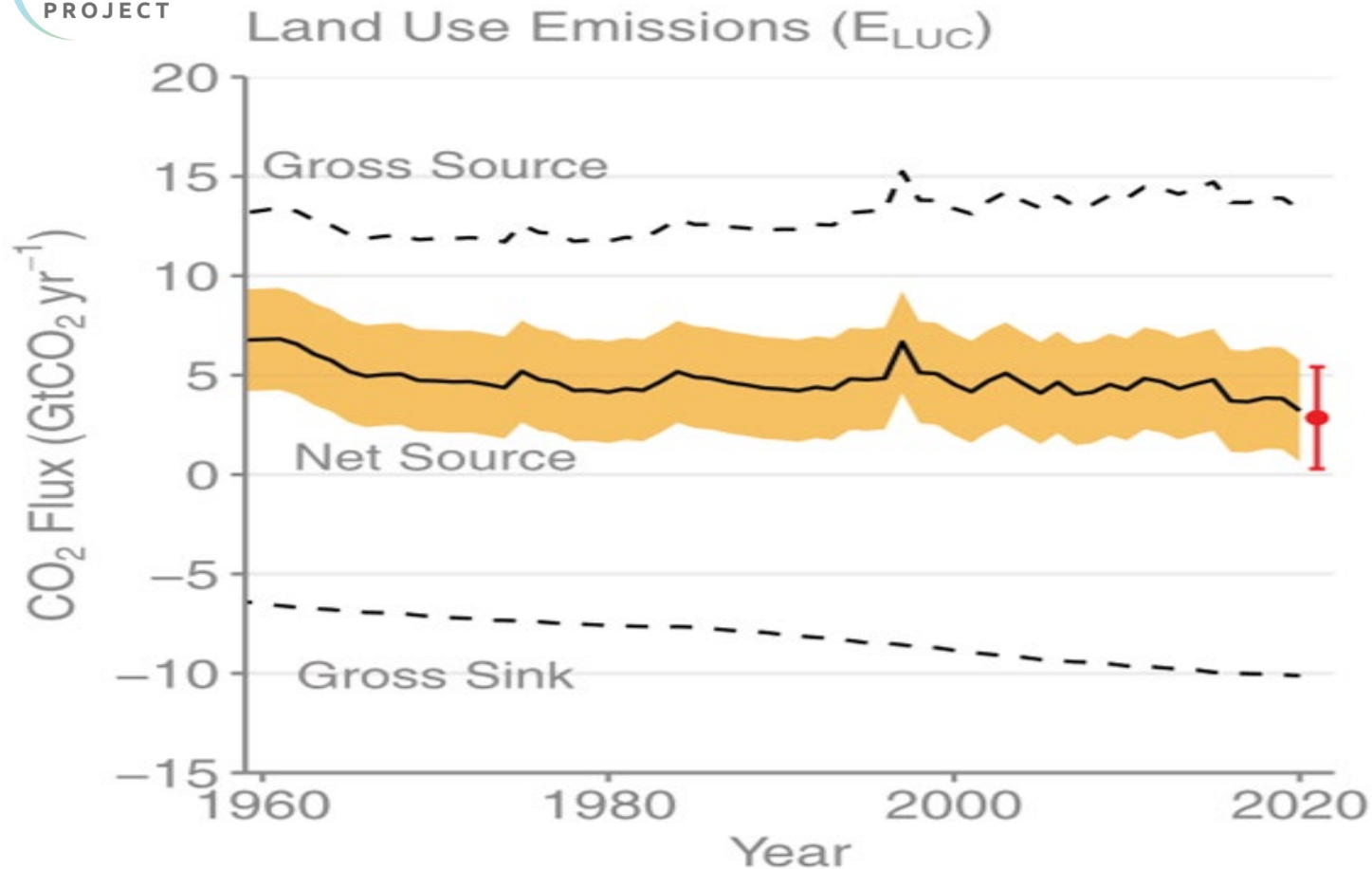
Budget Changes Over Time Averages for last decade (2011-2020)



Global fossil CO₂ emissions: 34.8 ± 2 GtCO₂ in 2020, 53% over 1990
2021 emissions: 36.4 ± 2 GtCO₂ [9.9 PgC], 4.9% higher than 2020



Budget Changes Over Time Averages for last decade (2011-2020)



1.1 ± 0.5 PgC/yr (4 GtCO₂)

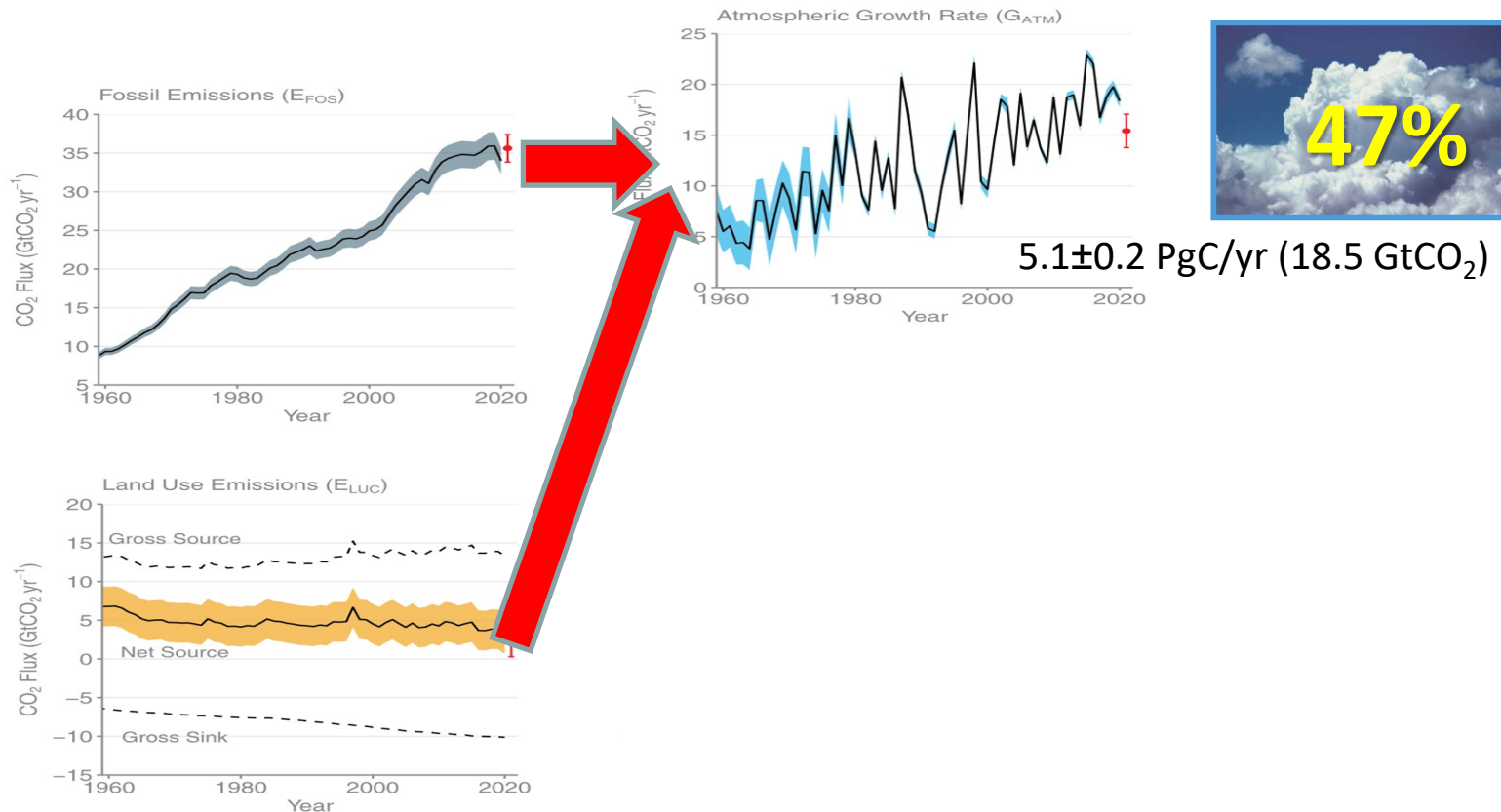
Budget Changes Over Time Averages for last decade (2011-2020)



9.5 ± 0.5 PgC/yr
(35 GtCO₂)



1.1 ± 0.5 PgC/yr
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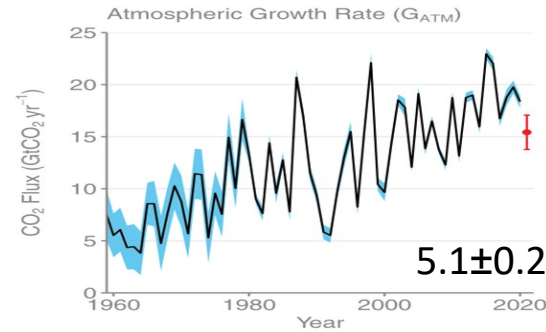
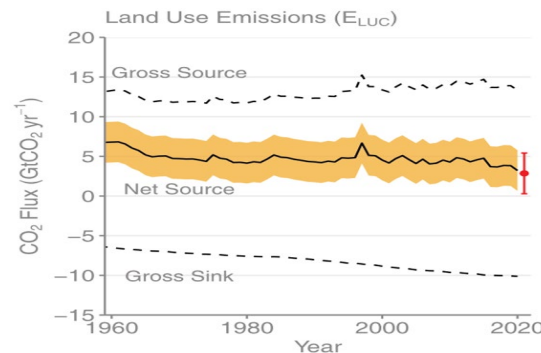
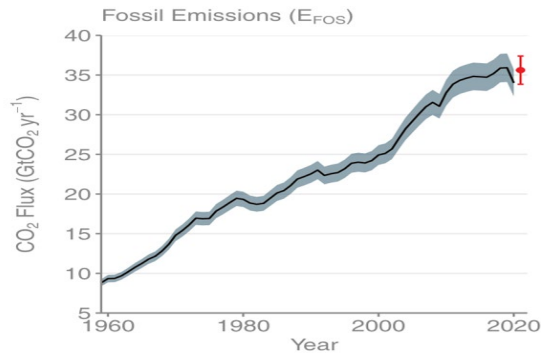
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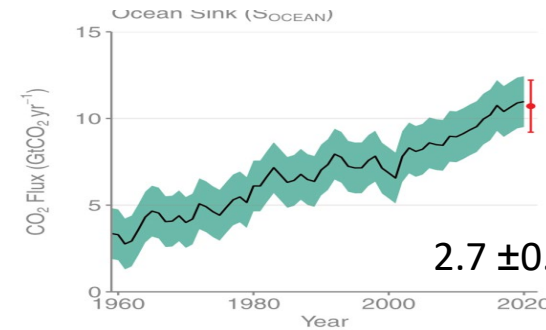
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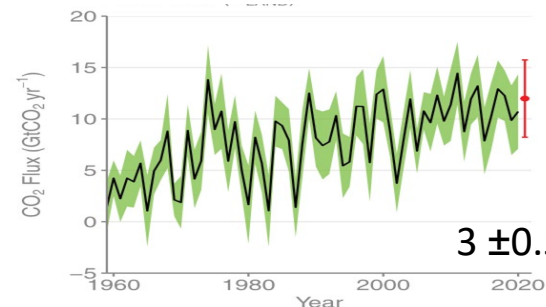
1.1 ± 0.5 PgC/yr
(4 GtCO₂)



5.1 ± 0.2 PgC/yr (18.5 GtCO₂)



2.7 ± 0.2 PgC/yr (10 GtCO₂)



3 ± 0.5 PgC/yr (11 GtCO₂)



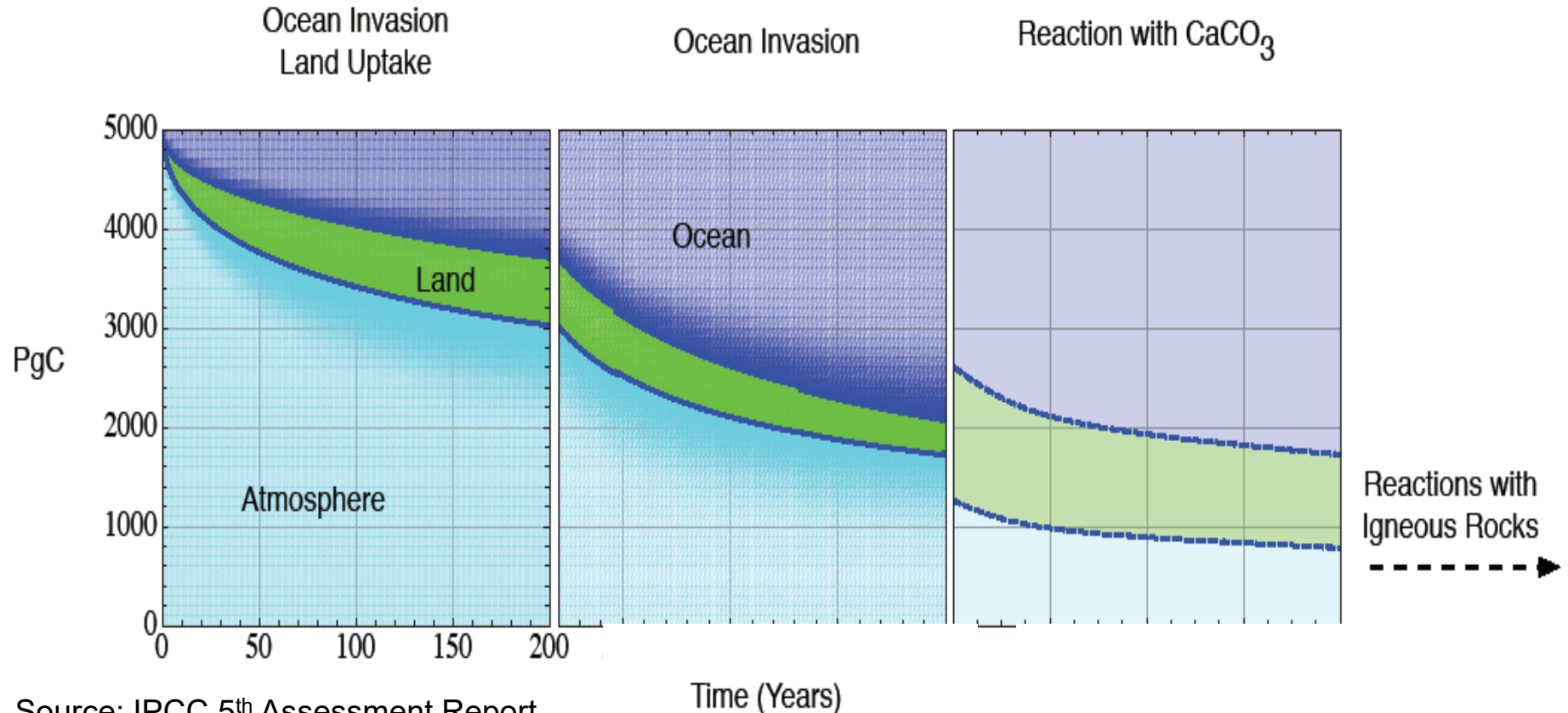
What are the long-term consequences of fossil fuel burning?

Hypothetical scenario: We instantly inject 5,000 Pg of carbon into the atmosphere

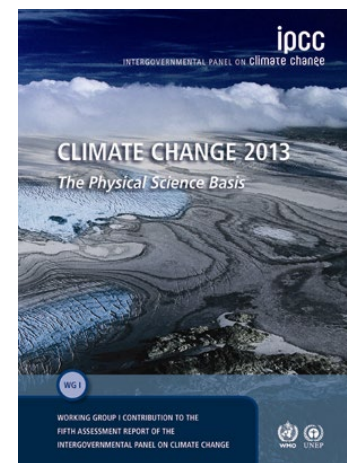
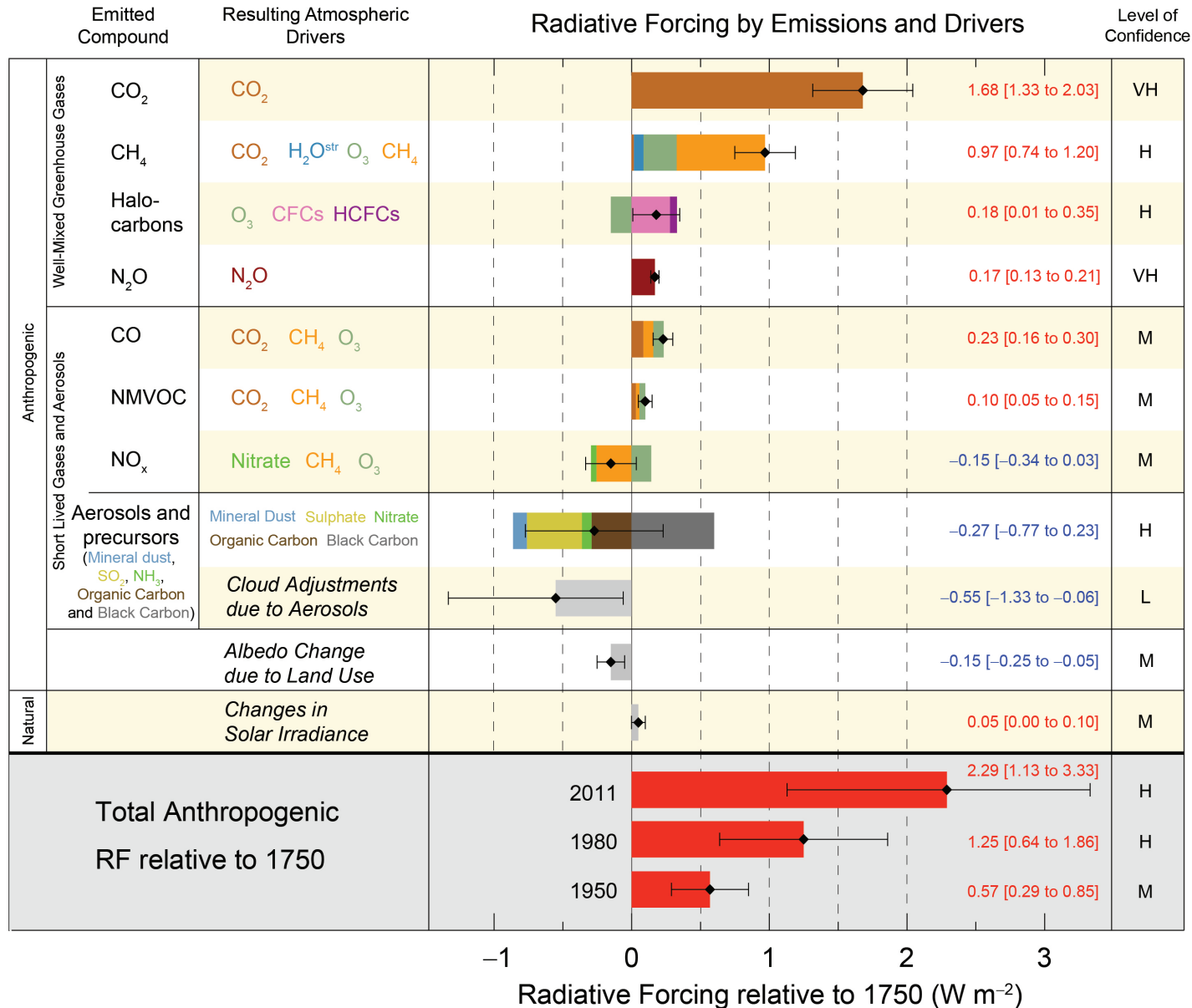
Time-scale of a few hundred years: primary storage is atmosphere with ocean and land absorption

Time-scale of a few thousand years: land sink maxed out and ocean becomes primary storage location

Time-scale of ten thousand years: ocean is primary storage location and mineral sink growing



What causes climate change?

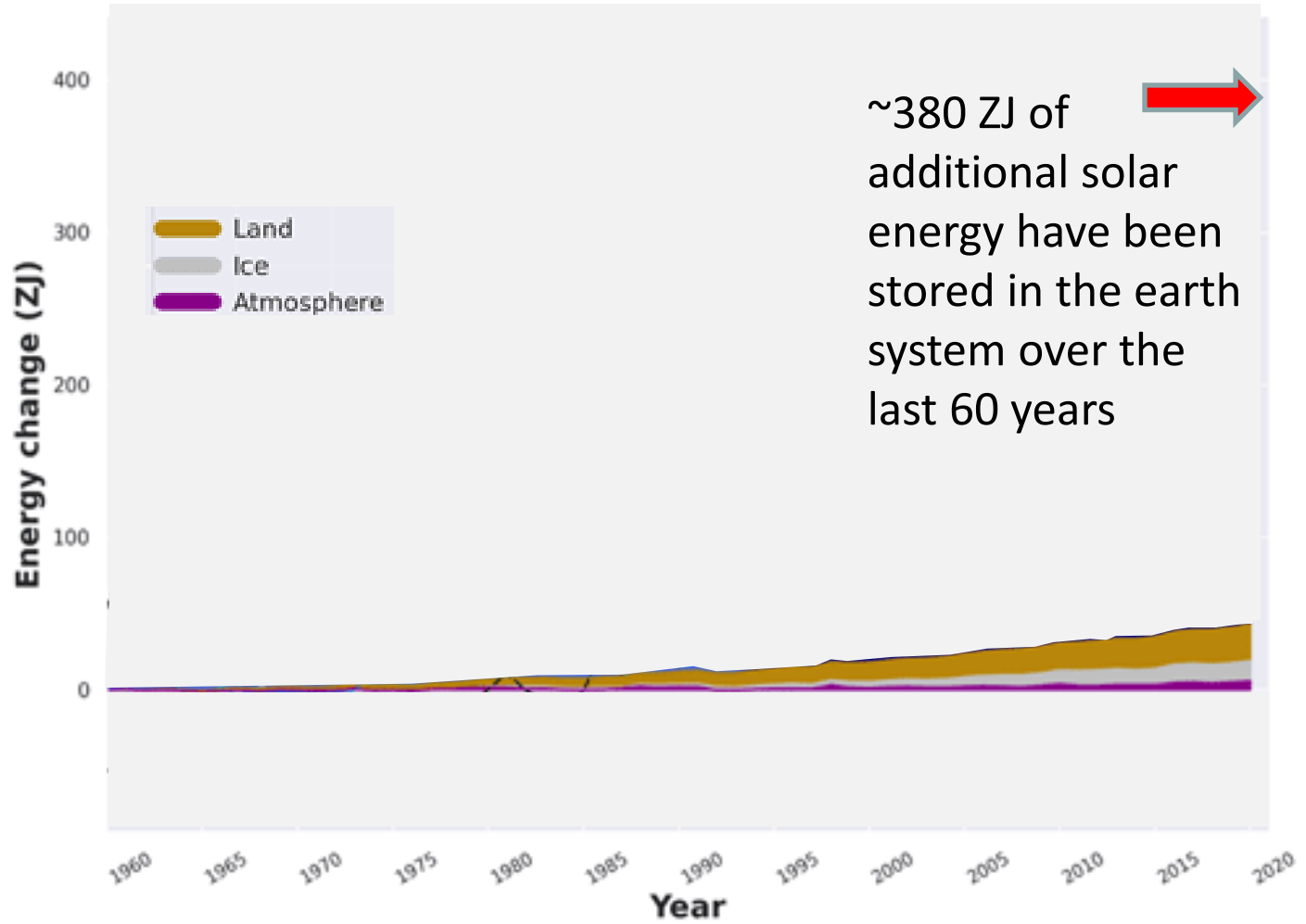


Global Energy Storage

Atmospheric warming accounts for about 1% of energy storage

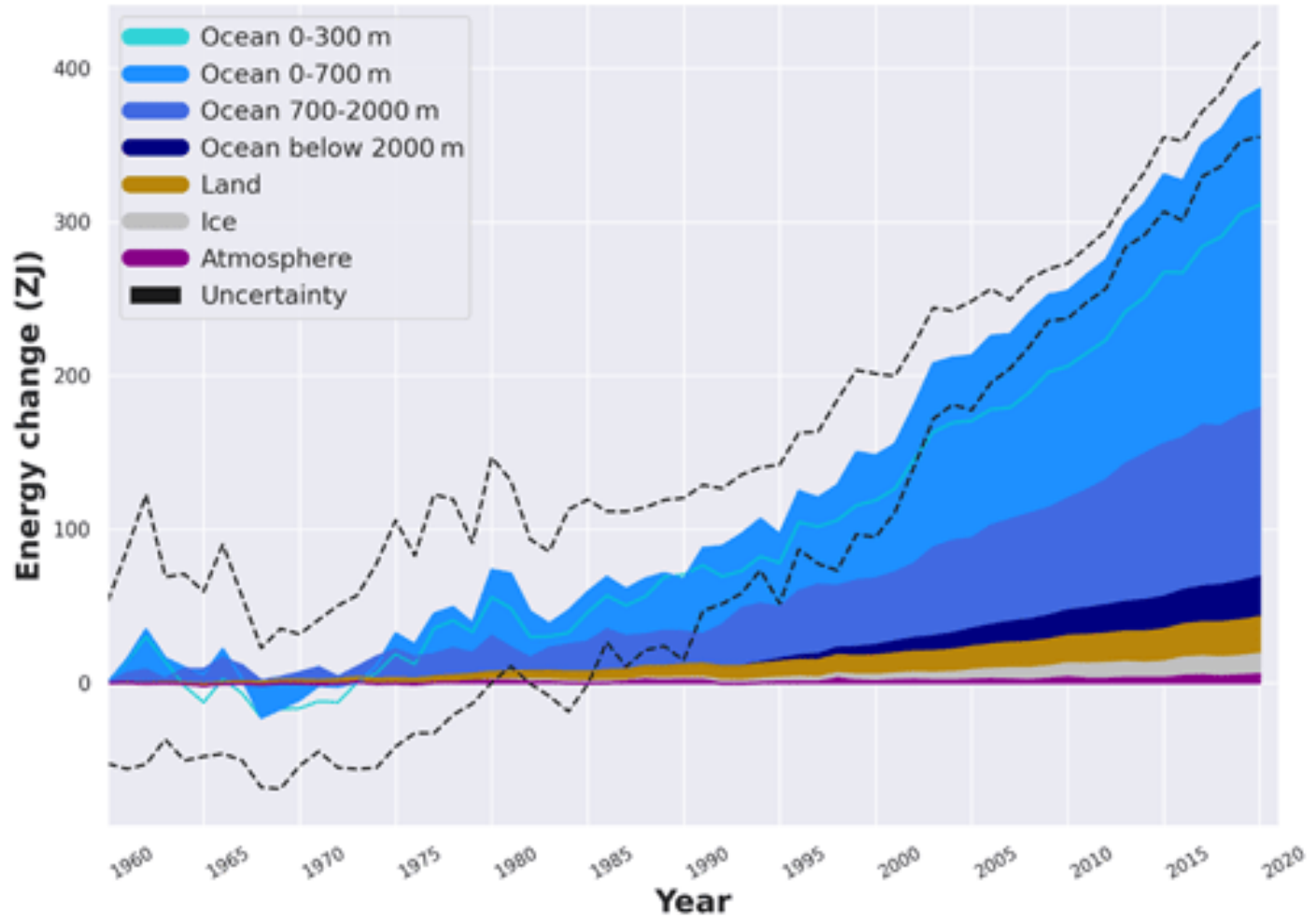
Melting Ice (including Arctic sea ice) accounts for ~3% of energy storage

Warming Land accounts for ~3% of energy storage



Global Energy Storage

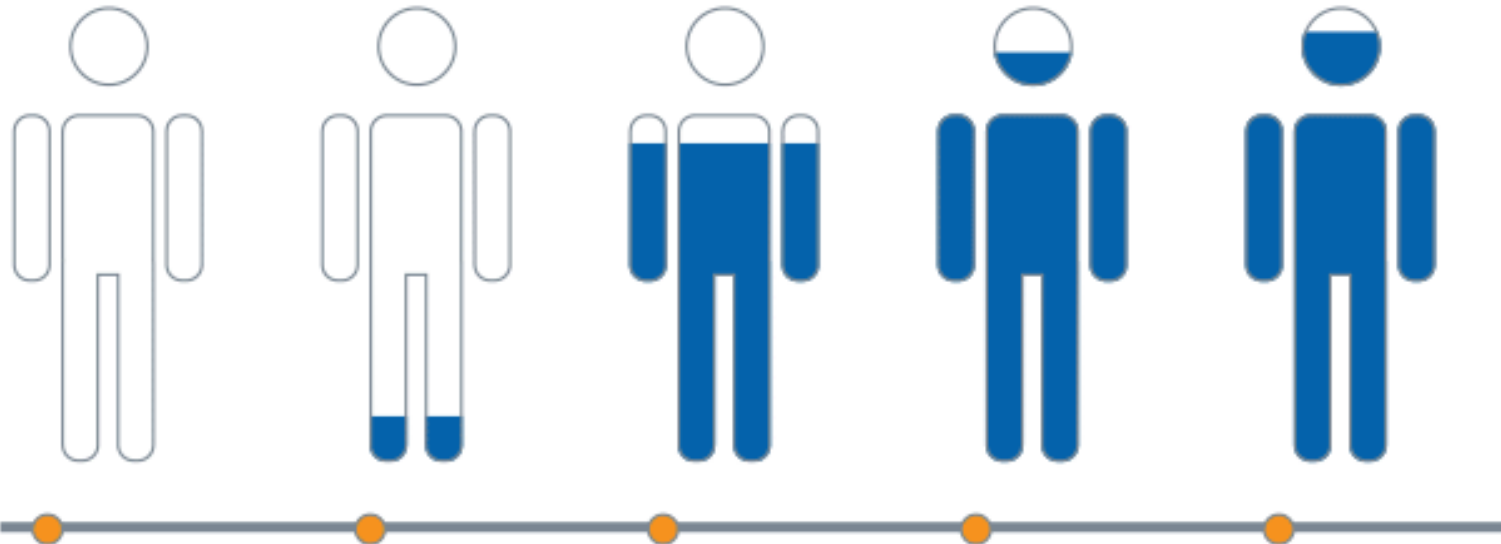
Ocean warming accounts for about ~93% of total energy storage



The heat we are putting in the ocean will be trapped there for many thousands of years

Increasing Confidence With Each Assessment

■ Amount of Human-caused Warming



1990

The report did not quantify the human contribution to global warming.

1995

"The balance of evidence suggests a **discernible** human influence on climate."

2001

Human-emitted greenhouse gases are **likely** (67-90% chance) responsible for more than half of Earth's temperature increase since 1951.

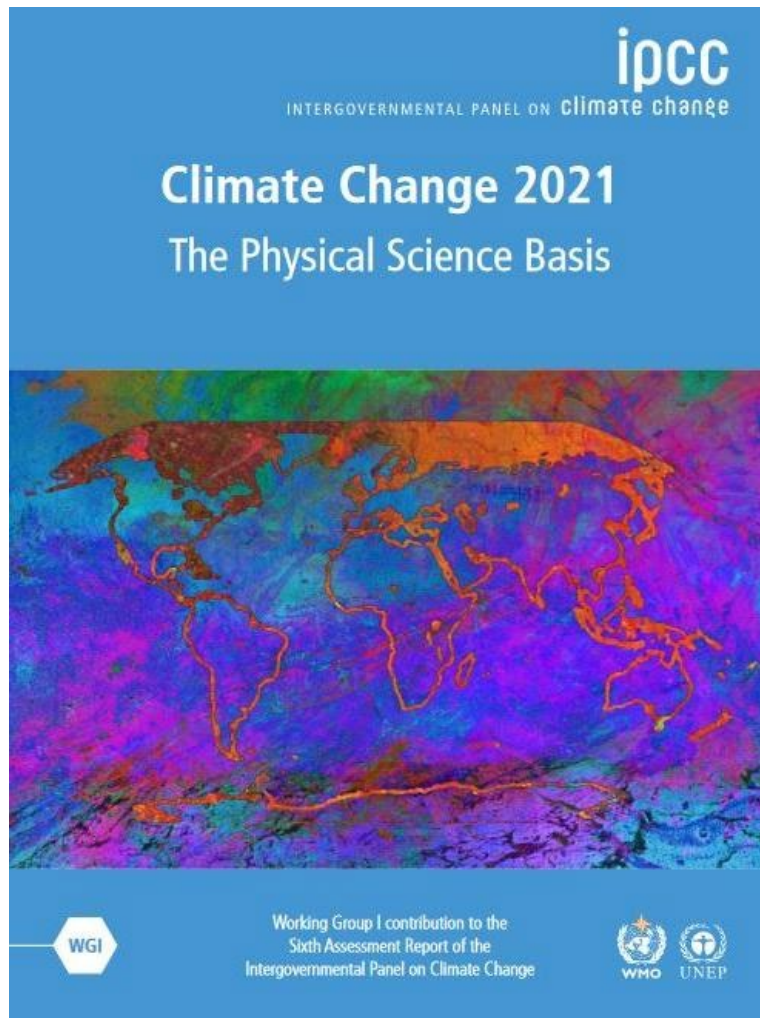
2007

Human-emitted greenhouse gases are **very likely** (at least 90% chance) responsible for more than half of Earth's temperature increase since 1951.

2013

Human-emitted greenhouse gases are **extremely likely** (at least 95% chance) responsible for more than half of Earth's temperature increase since 1951.

IPCC 6th Assessment Report



A.1 It is unequivocal that human influence has warmed the atmosphere, ocean and land. Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred.

A.2 The scale of recent changes across the climate system as a whole and the present state of many aspects of the climate system are unprecedented over many centuries to many thousands of years.

A.3 Human-induced climate change is already affecting many weather and climate extremes in every region across the globe. Evidence of observed changes in extremes such as heatwaves, heavy precipitation, droughts, and tropical cyclones, and, in particular, their attribution to human influence, has strengthened since the Fifth Assessment Report (AR5).

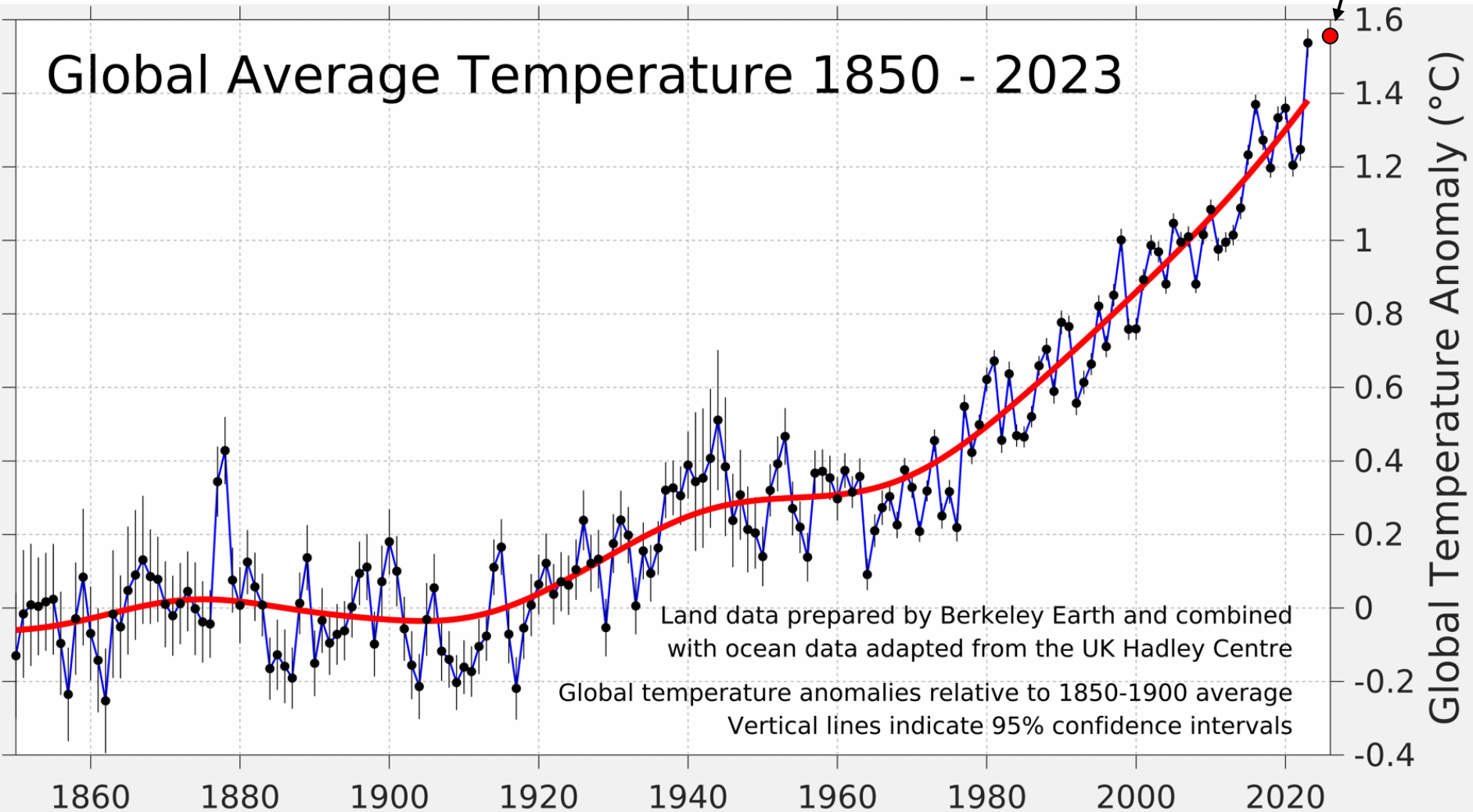
A.4 Improved knowledge of climate processes, paleoclimate evidence and the response of the climate system to increasing radiative forcing gives a best estimate of equilibrium climate sensitivity of 3°C, with a narrower range compared to AR5.

2023 and 2024 Were the Warmest Years on Record, surpassing the 1.5°C goal

Global Average Temperature 1850 - 2023

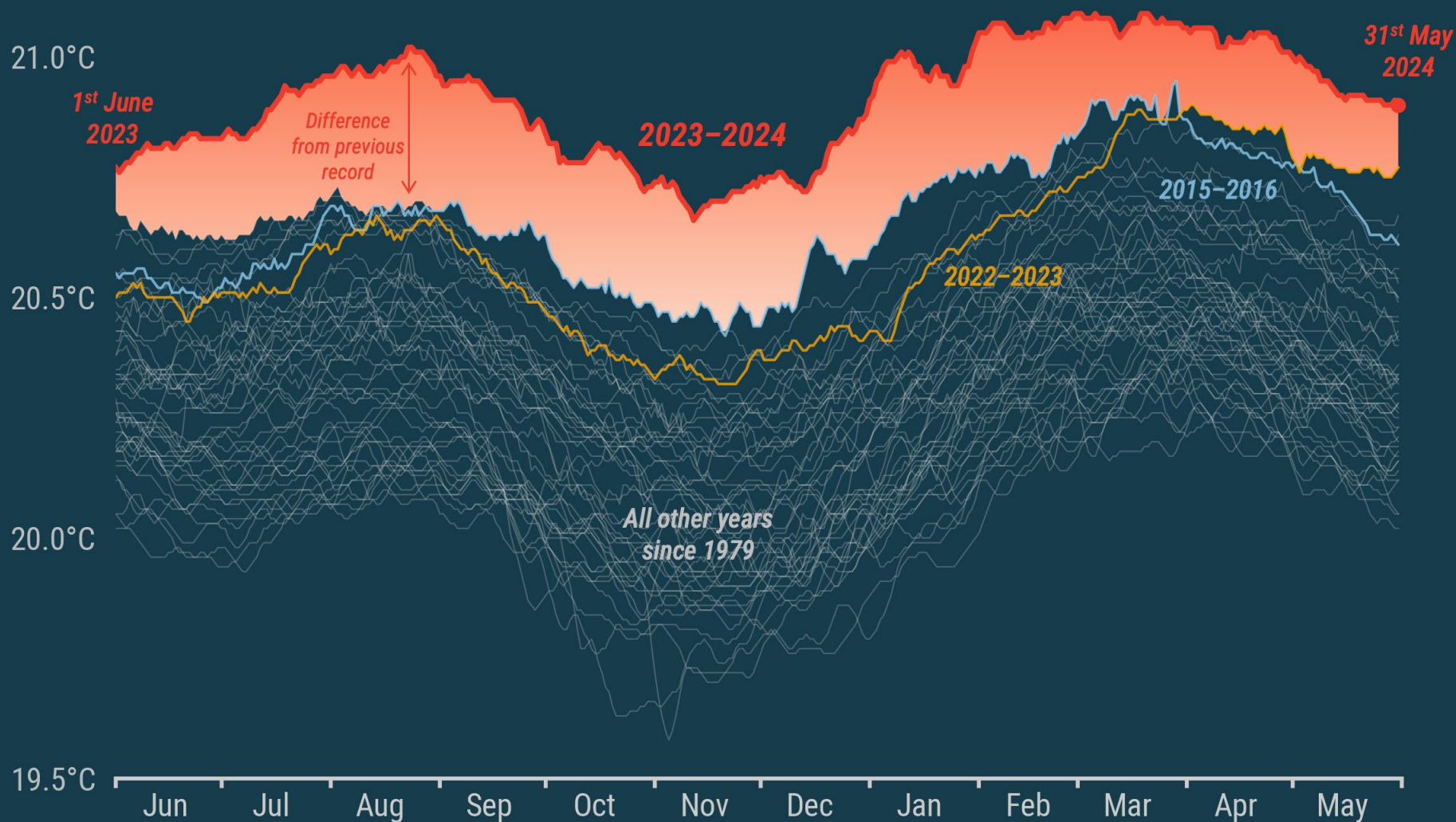
2024

Global Temperature Anomaly (°C)



Daily sea surface temperature for 60°S-60°N

Data: ERA5 1979–2024 • Credit: C3S/ECMWF



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